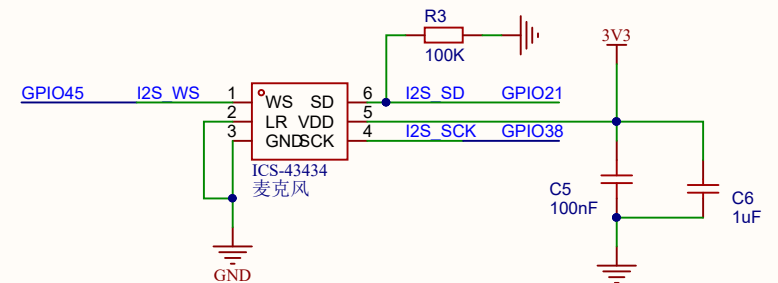
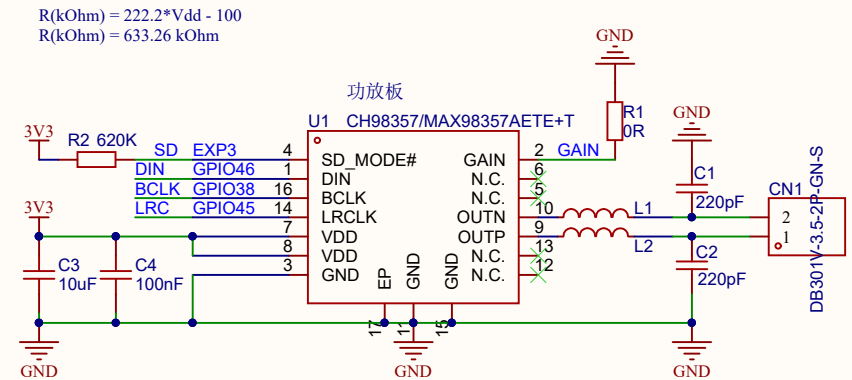


## Connection diagram

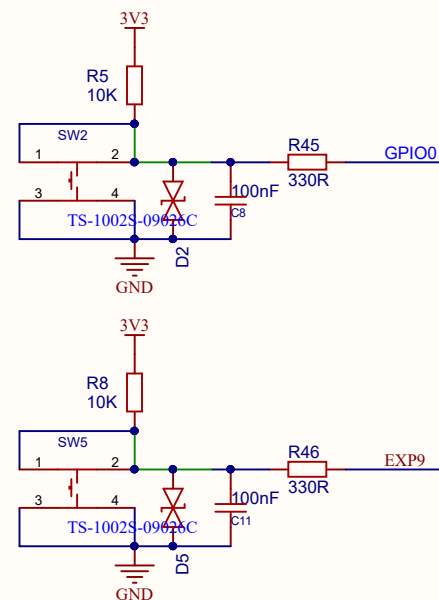
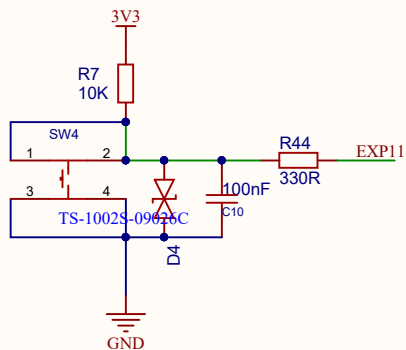
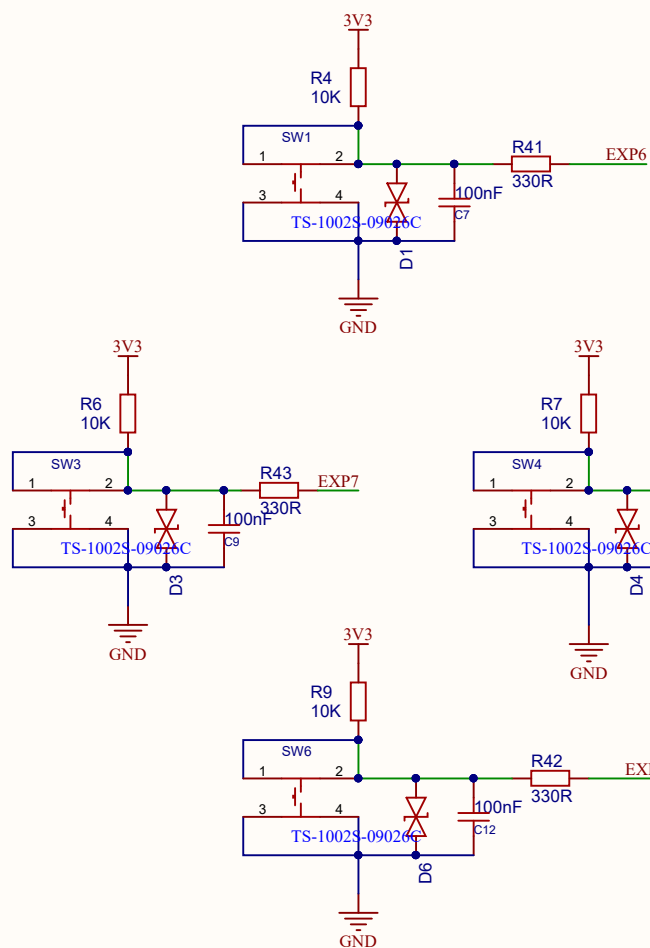
The speaker is connected to the ESP32-S3 as follows:

| Speaker | ESP32-S3 |
|---------|----------|
| SCK     | GPIO38   |
| WS      | GPIO45   |
| DOUT    | GPIO46   |
| SD      | EXP3     |

| Microphone | ESP32-S3 |
|------------|----------|
| SCK        | GPIO38   |
| WS         | GPIO45   |
| DIN        | GPIO21   |

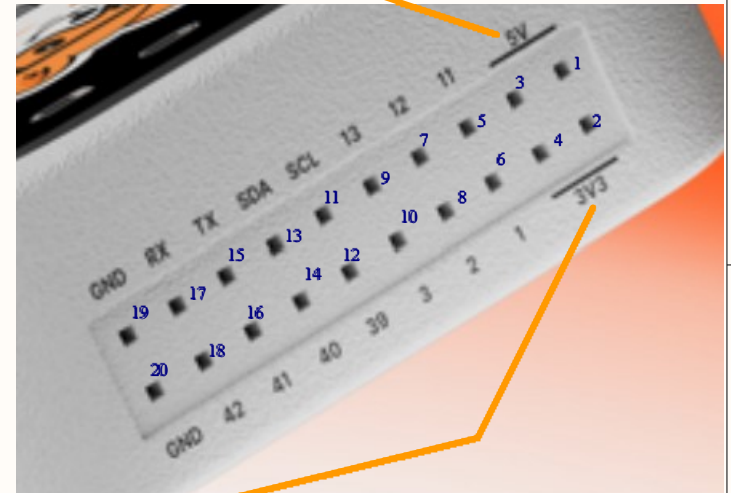
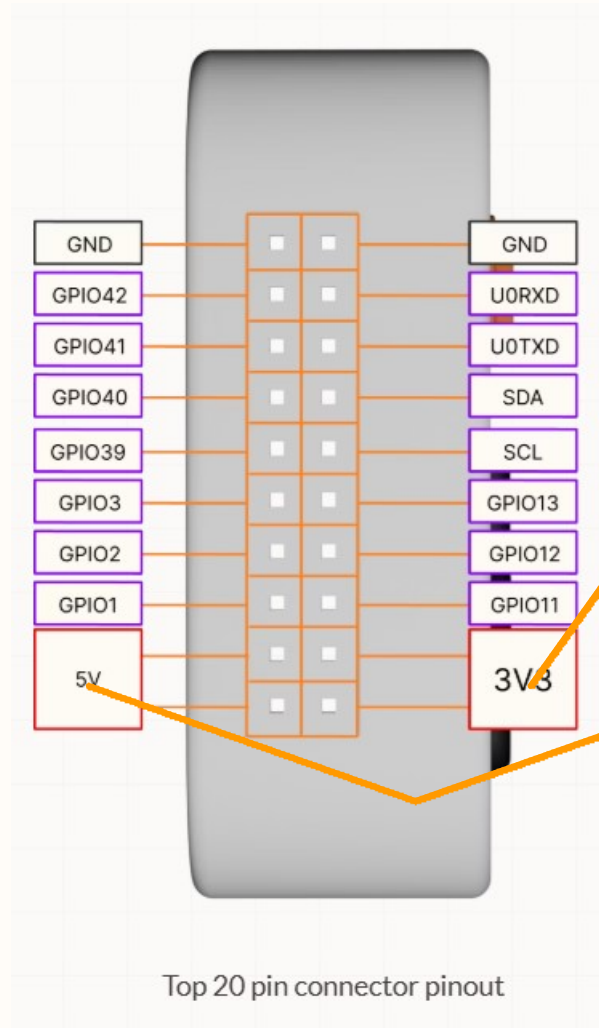


|            |                               |           |
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| Title      |                               |           |
| Size<br>A4 | Number                        | Revision  |
| Date:      | 2026/5/11                     | Sheet of  |
| File:      | E:\0 project\...\Audio.SchDoc | Drawn By: |



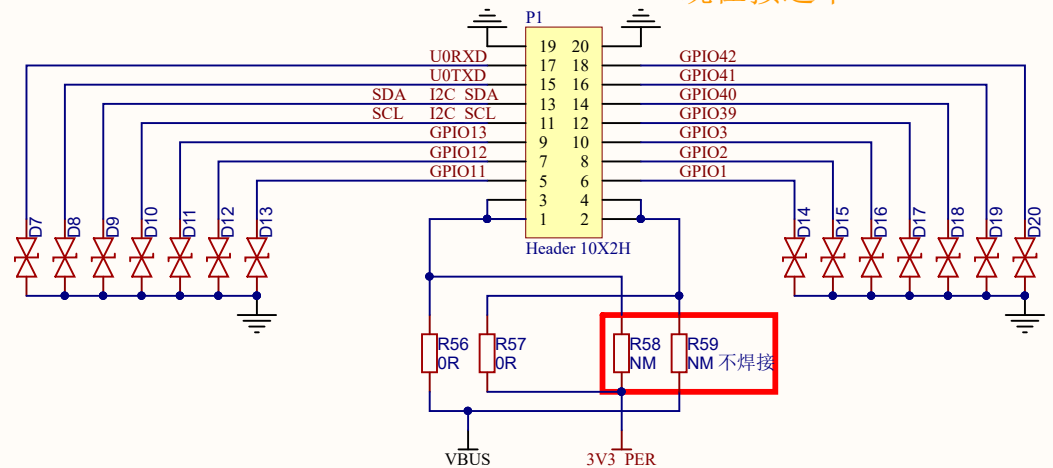
轻触开关

| Title |                                |           |
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| Size  | Number                         | Revision  |
| A4    |                                |           |
| Date: | 2026/5/11                      | Sheet of  |
| File: | E:\0 project\...\Button.SchDoc | Drawn By: |



这两个交叉了。不知道用哪个？

现在按这个



|       |                                 |           |
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| Title |                                 |           |
| Size  | Number                          | Revision  |
| A4    |                                 |           |
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| File: | E:\0 project\.\Connector.SchDoc | Drawn By: |

CO5300

ESP32-S3

Chip Select

GPIO9

Clock

GPIO17

Data 0

GPIO15

Data 1

GPIO14

Data 2

GPIO16

Data 3

GPIO10

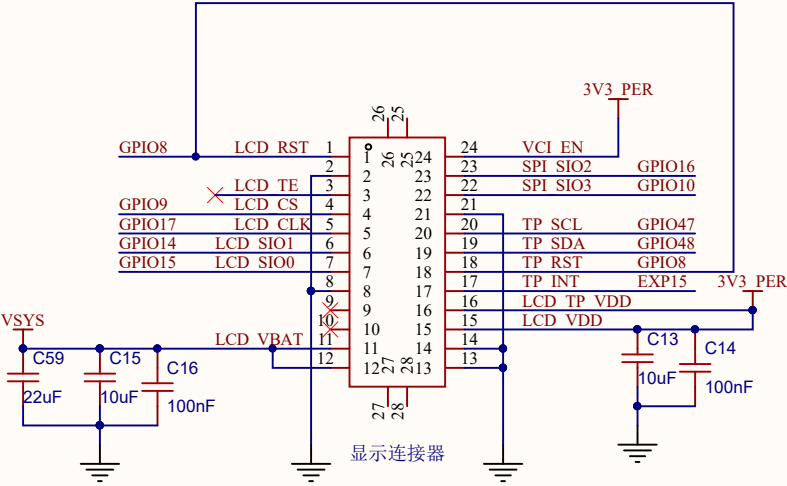
Reset

GPIO8

CO5300  
The QuadSPI bus is of the same family as the SPI bus, but has double the bandwidth.

CST820  
The touch driver has the address 0x15 on the I2C bus.

v0.9修改 1, 加大电容; 2, vSYS



| Title |                                 |           |
|-------|---------------------------------|-----------|
| Size  | Number                          | Revision  |
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| File: | E:\0 project\...\Display.SchDoc | Drawn By: |

|      |          |
|------|----------|
| IMU  | ESP32-S3 |
| SDA  | GPIO48   |
| SCL  | GPIO47   |
| INT1 | EXP13    |
| INT2 | EXP12    |

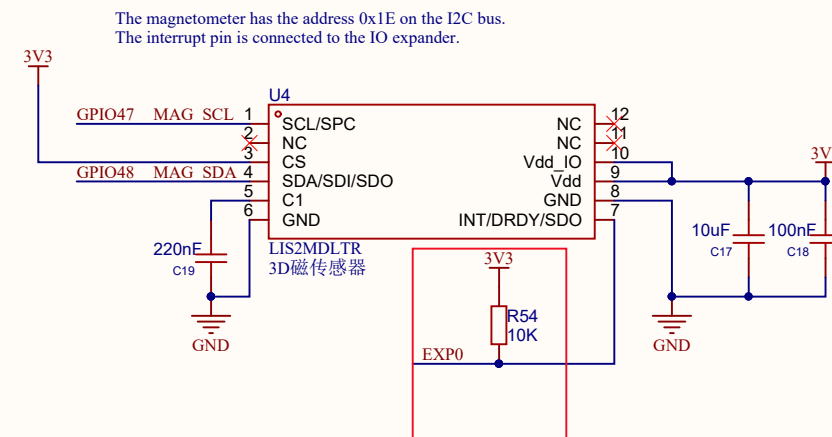
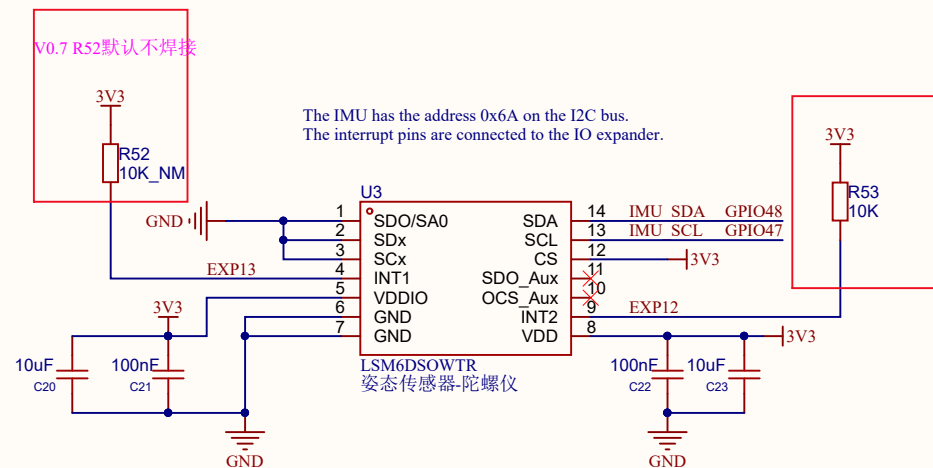
💡 The IMU has the address 0x6A on the I2C bus.

❗ The interrupt pins are connected to the IO expander.

## 3-axis magnetometer

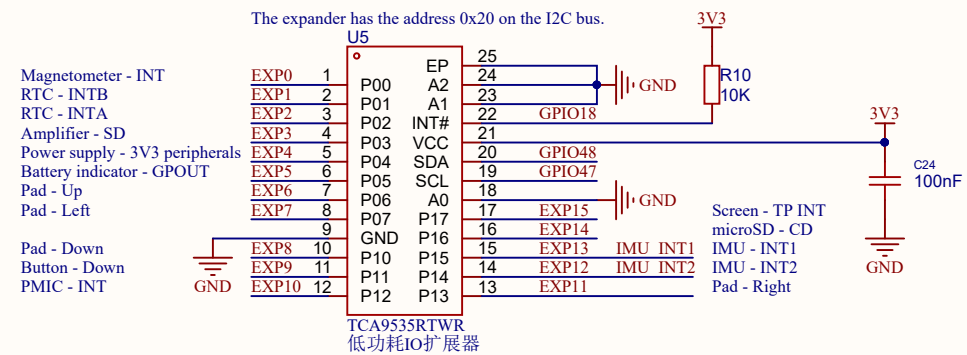
The magnetometer is connected in the same way to the I2C

|              |          |
|--------------|----------|
| Magnetometer | ESP32-S3 |
| SDA          | GPIO48   |
| SCL          | GPIO47   |
| INT1         | EXP0     |

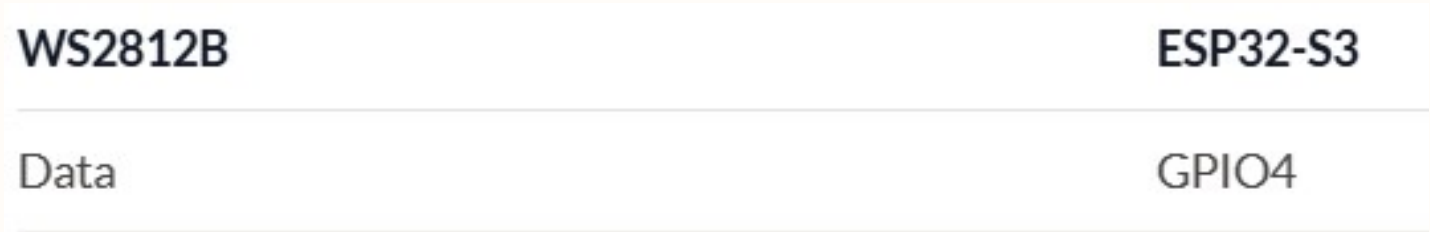


|       |                           |           |
|-------|---------------------------|-----------|
| Title |                           |           |
| Size  | Number                    | Revision  |
| A4    |                           |           |
| Date: | 2026/5/11                 | Sheet of  |
| File: | E:\0 project\..IMU.SchDoc | Drawn By: |

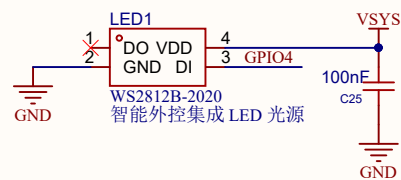
| Signal                         | Expander |
|--------------------------------|----------|
| Magnetometer - INT             | EXP0     |
| RTC - INTB                     | EXP1     |
| RTC - INTA                     | EXP2     |
| Amplifier - SD                 | EXP3     |
| Power supply - 3V3 peripherals | EXP4     |
| Battery indicator - GPOUT      | EXP5     |
| Pad - Up                       | EXP6     |
| Pad - Left                     | EXP7     |
| Pad - Down                     | EXP8     |
| Button - Down                  | EXP9     |
| PMIC - INT                     | EXP10    |
| Pad - Right                    | EXP11    |
| IMU - INT2                     | EXP12    |
| IMU - INT1                     | EXP13    |
| microSD - CD                   | EXP14    |
| Screen - TP INT                | EXP15    |



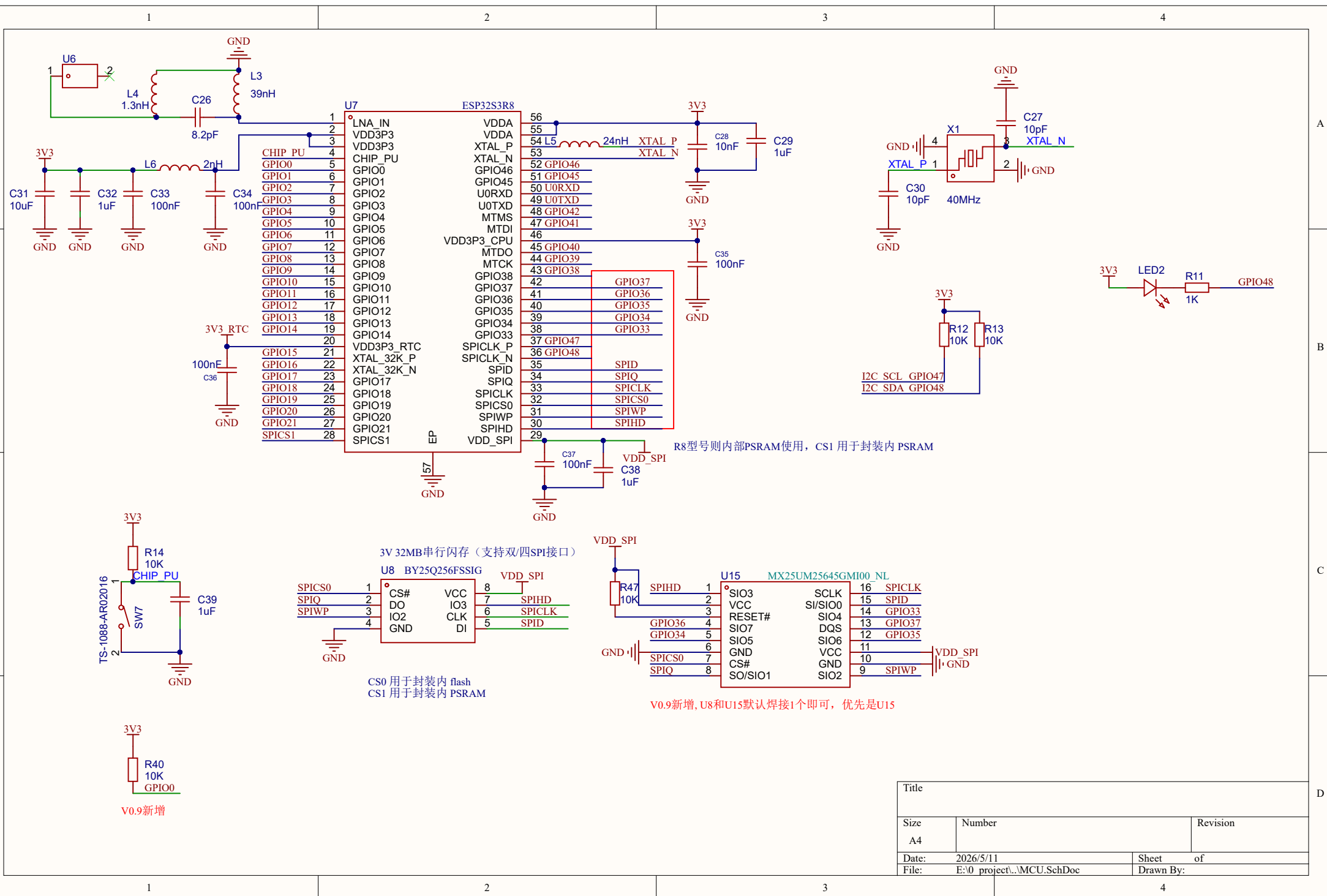
|            |                                   |           |
|------------|-----------------------------------|-----------|
| Title      |                                   |           |
| Size<br>A4 | Number                            | Revision  |
| Date:      | 2026/5/11                         | Sheet of  |
| File:      | E:\0 project\..MO Expander.SchDoc | Drawn By: |



1-Wire is a communication protocol that allows data communication through a single pin.

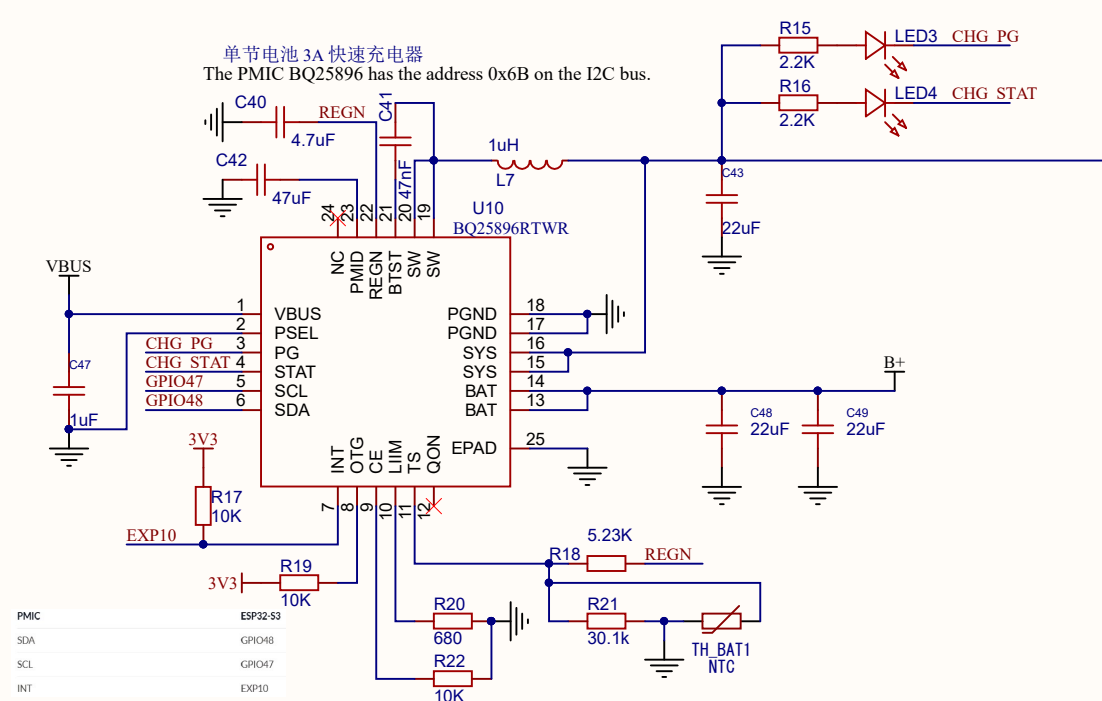


| Title |                             |           |
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| Date: | 2026/5/11                   | Sheet of  |
| File: | E:\0 project\...\LED.SchDoc | Drawn By: |





单节电池 3A 快速充电器  
The PMIC BQ25896 has the address 0x6B on the I2C bus.



OTG拉高，软件OTG\_CONFIG = 1，CE接GND，上电VBUS输出5V

插 USB Type-C插入自动停 OTG，正常充电。

SW8  
MSKT-12D18A贴片小拨动

1A 电流：  
(1) ESP32-S3 主控  
(2) IO expander IO 口扩展器  
(3) RTC 时间芯片  
(4) Audio amplifier 音频放大器  
(5) Microphone 麦克风  
(6) 6-axis IMU 姿态传感器  
(7) 3-axis magnetometer 磁力计

3V3 regulator for internal components

ESP32-S3 RTC

2A 电流：  
(1) Screen 屏幕  
(2) microSD 内存卡  
(3) 3V3 bus of the upper connector 顶部20针连接器

$$V_{out} = 0.6 \times (1 + R_{10}/R_{12}) = 3.315V$$

3V3 regulator for peripherals

3V3 regulator for peripherals  
This regulator, which supports up to 2A of current, can be activated or deactivated to power the following peripherals of the Kode Dot:  
(1) Screen  
(2) microSD  
(3) 3V3 bus of the upper connector  
When the Kode Dot is in suspend mode, this regulator is deactivated and disconnects the power to the peripherals.

Fuel Gauge  
SDA  
SCL  
GPOUT

ESP32-S3  
GPIO48  
GPIO47  
EXP5

500mAh 503035 3.7V 1.05WH

通过 bq27220 器件进行电池电量监测时，只需将 C3 与 C1 连接至可拆卸电池组或嵌入式电池电路即可

The fuel gauge has the address 0x55 on the I2C bus.

| Title |                           |           |
|-------|---------------------------|-----------|
| Size  | Number                    | Revision  |
| A4    |                           |           |
| Date: | 2026/5/11                 | Sheet of  |
| File: | E:\0 project\Power.SchDoc | Drawn By: |

External RTC

ESP32-S3

SDA

GPIO48

SCL

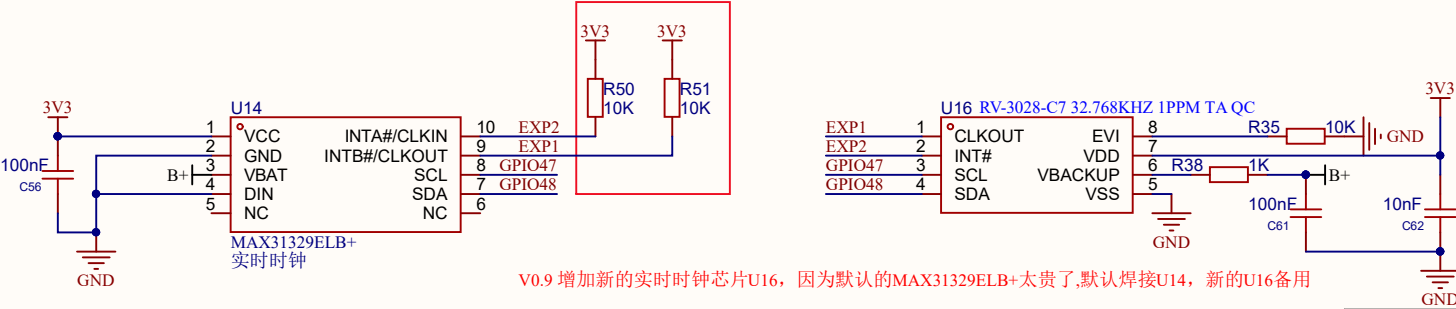
GPIO47

INTA

EXP2

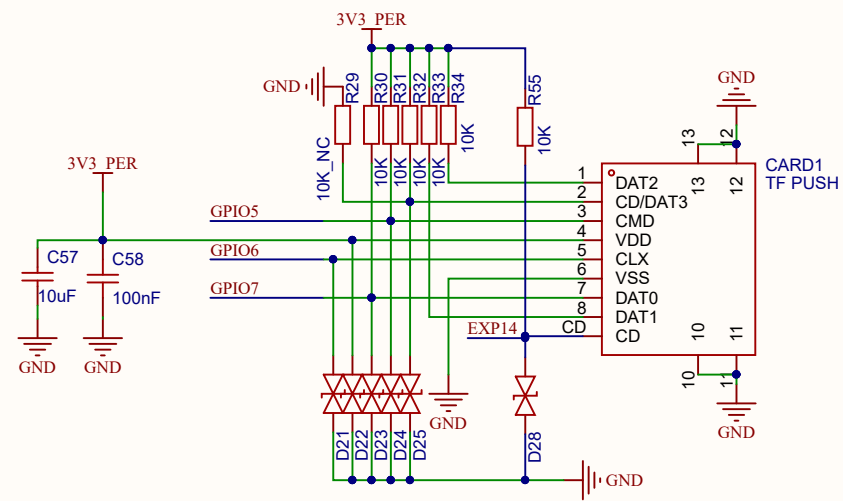
INTB

EXP1



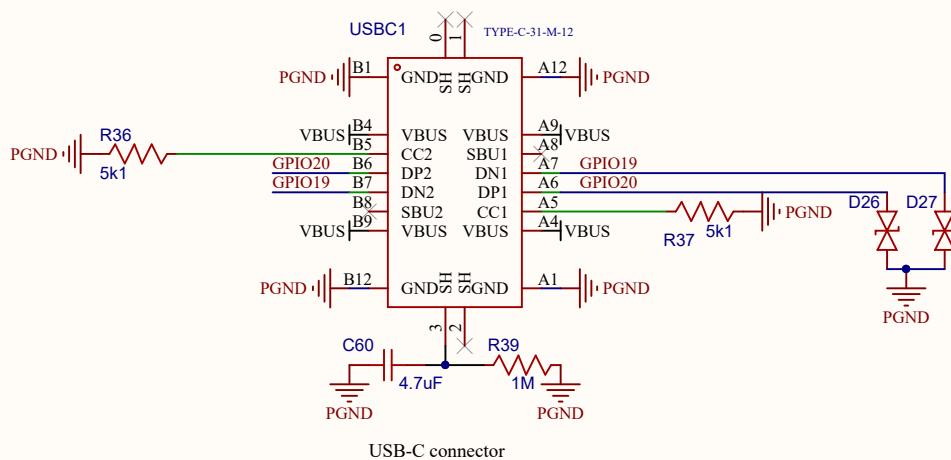
v0.9 增加新的实时时钟芯片U16，因为默认的MAX31329ELB+太贵了,默认焊接U14，新的U16备用

| Title |                           |           |
|-------|---------------------------|-----------|
| Size  | Number                    | Revision  |
| A4    |                           |           |
| Date: | 2026/5/11                 | Sheet of  |
| File: | E:\0 project\..RTC.SchDoc | Drawn By: |

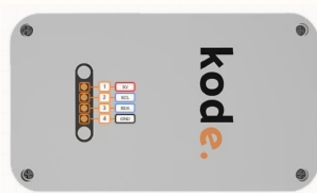
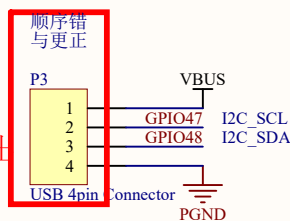


| MicroSD       | ESP32-S3 |
|---------------|----------|
| Command       | GPIO5    |
| Clock         | GPIO6    |
| Data          | GPIO7    |
| Card Detected | EXP14    |

| Title |                               |           |          |
|-------|-------------------------------|-----------|----------|
| Size  | Number                        |           | Revision |
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| Date: | 2026/5/11                     | Sheet     | of       |
| File: | E:\0 project\..sd card.SchDoc | Drawn By: |          |



V0.9新增, 磁吸焊盘排线引出



|       |                            |           |
|-------|----------------------------|-----------|
| Title |                            |           |
| Size  | Number                     | Revision  |
| A4    |                            |           |
| Date: | 2026/5/11                  | Sheet of  |
| File: | E:\0 project\..USBC.SchDoc | Drawn By: |

